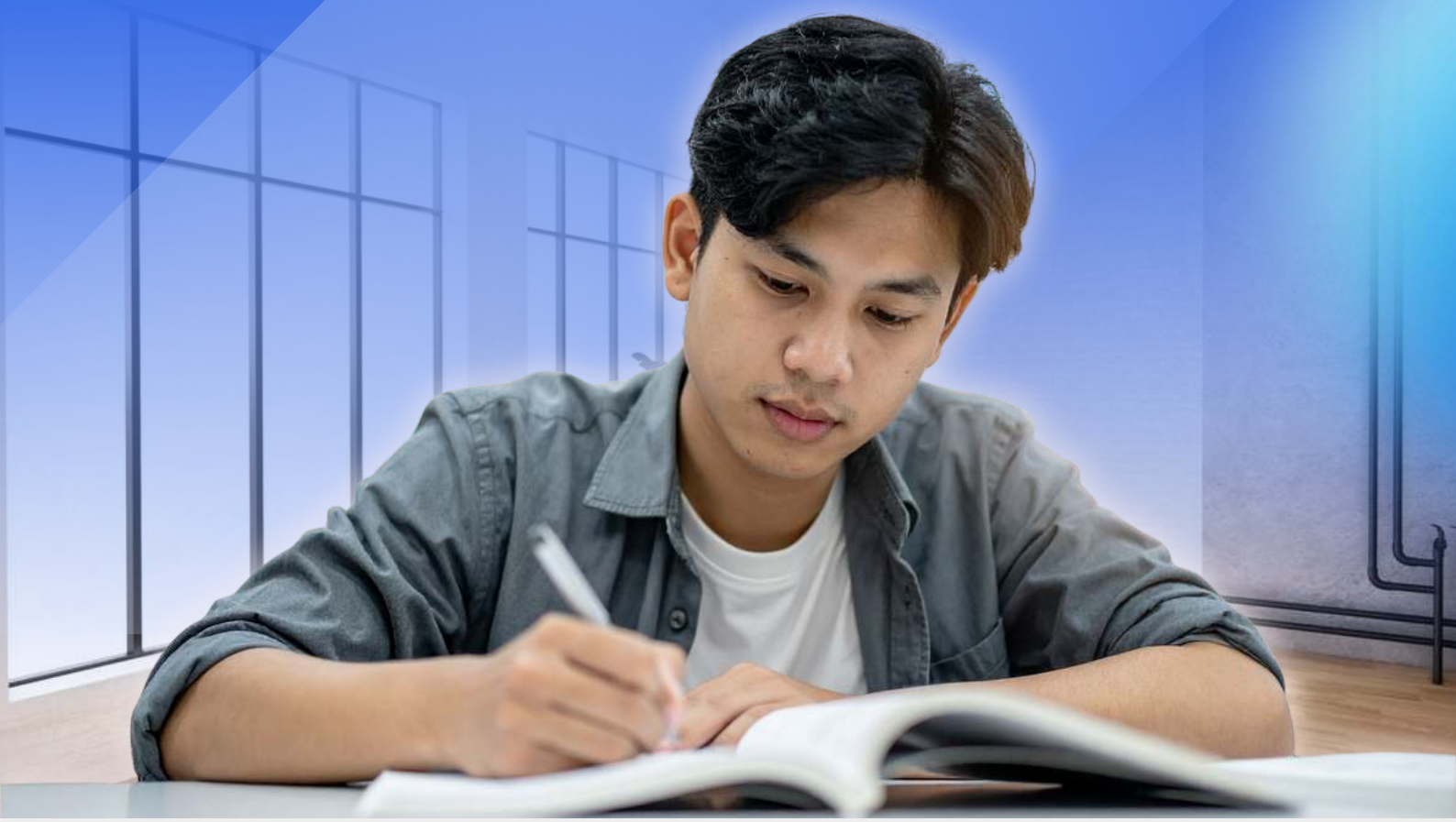
Lenterago

TRY OUT #7

SAT-Math

61007

54 QUESTIONS



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MODUL 1 & 2

22 QUESTIONS

DIRECTIONS

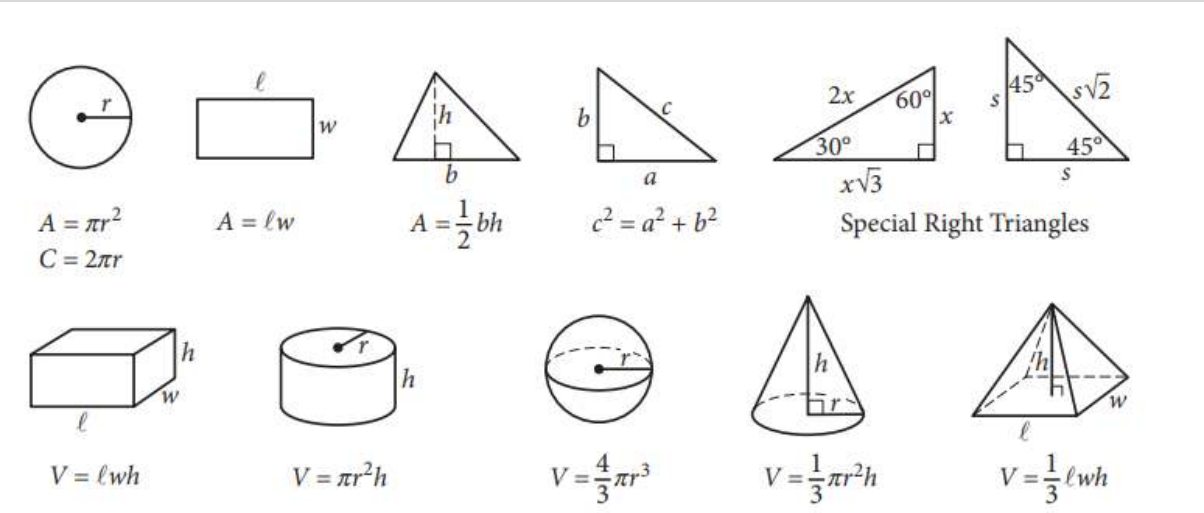
The questions in this section address a number of important math skills.
Use of a calculator is permitted for all questions.

NOTES

Unless otherwise indicated:

- All variables and expressions represent real numbers.
- Figures provided are drawn to scale.
- All figures lie in a plane.
- The domain of a given function *f* is the set of all real numbers *x* for which *f(x)* is a real number

REFERENCE



The number of degrees of arc in a circle is 360.

The number of radians of arc in a circle is 2p.

The sum of the measures in degrees of the angles of a triangle is 180.

0 8 2 1 7 0 2 7 6 0 7 6

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Module 1 & 2

For multiple-choice questions, solve each problem, choose the correct answer from the choices provided, and then circle your answer in this book. Circle only one answer for each question. If you change your mind, completely erase the circle. You will not get credit for questions with more than one answer circled, or for questions with no answers circled.

For student-produced response questions, solve each problem and write your answer next to or under the question in the test book as described below.

- Once you've written your answer, circle it clearly. You will not receive credit for anything written outside the circle, or for any questions with more than one circled answer.
- If you find more than one correct answer, write and circle only one answer.
- Your answer can be up to 5 characters for a positive answer and up to 6 characters (including the negative sign) for a negative answer, but no more.
- If your answer is a fraction that is too long (over 5 characters for positive, 6 characters for negative), write the decimal equivalent.
- If your answer is a decimal that is too long (over 5 characters for positive, 6 characters for negative), truncate it or round at the fourth digit.
- If your answer is a mixed number (such as 3 1/2), write it as an improper fraction (7/2) or its decimal equivalent (3.5).
- Don't include symbols such as a percent sign, comma, or dollar sign in your circled answer.

0 8 2 1 7 0 2 7 6 0 7 6

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NO	MATERI	SUB MATERI
1	Algebra	Linear Equation

If $4x - 28 = -24$ what is the value $14 - 2x$?

A. -24
B. -12
C. 6
D. 12

2	Problem Solving and Data Analysis	Percent

The result of increasing the quantity x by 400% is resulting 80. What is the value of x?

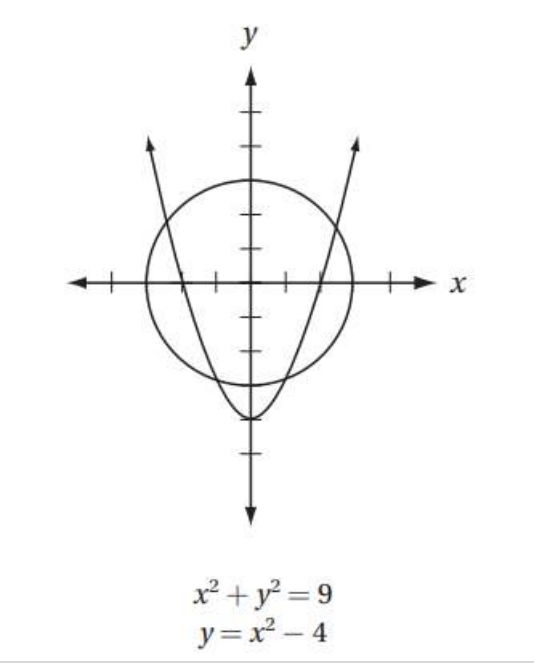
A. 12
B. 14
C. 16
D. 20

3	Algebra	Linear Equation & Functions

Tera deposited a fixed amount into her bank account each mount. The function $f(t) = 100 + 20t$ give the amount , in dollars, in Tera's bank account after t monthly deposits. What is the best interpretation of 20 in this context?

A. After 1 month deposit, the amount of Tera's bank account was $\$20$
B. Before Tera made any monthly deposits, the amount in her bank account was $\$20$
C. With each monthly deposit, the amount in Tera's bank account increased by $\$20$
D. Tera made a total 25 monthly deposits

4	Algebra	System of Equation



A system of two equation and their graph in the xy-plane are shown above. How many solutions does the system have?

A. None
B. Four
C. Six
D. Seven

0 8 2 1 7 0 2 7 6 0 7 6

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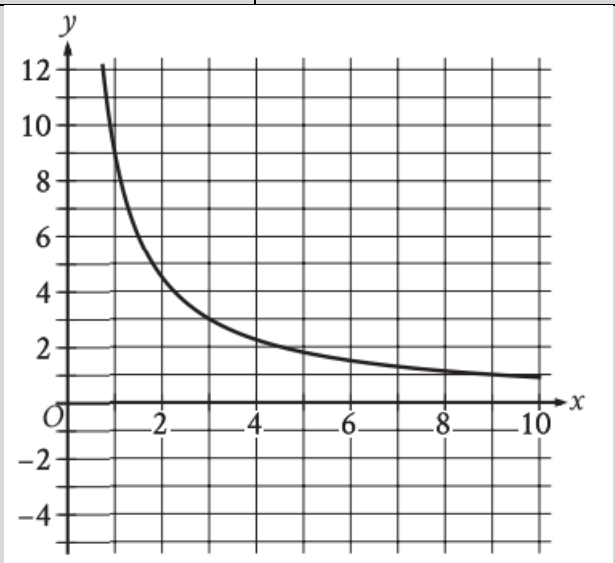
5	Algebra	System of Equation + Solution Properties	

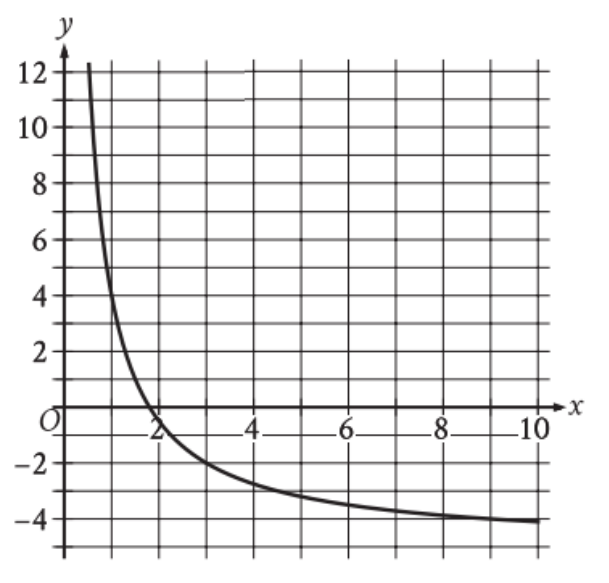
$$\frac{1}{2}x = \frac{1}{3}y + \frac{1}{10}$$
$$6x - 4y = k$$

For what value of k will the system of equation above have at least one solution?

6	Algebra	Functions	



The graph of the rational function f is shown, where $y = f(x)$ and $x \geq 0$. Which of the following is the graph of $h(x) = f(x) - 5$, where and $x \geq 0$?

A. 

0 8 2 1 7 0 2 7 6 0 7 6

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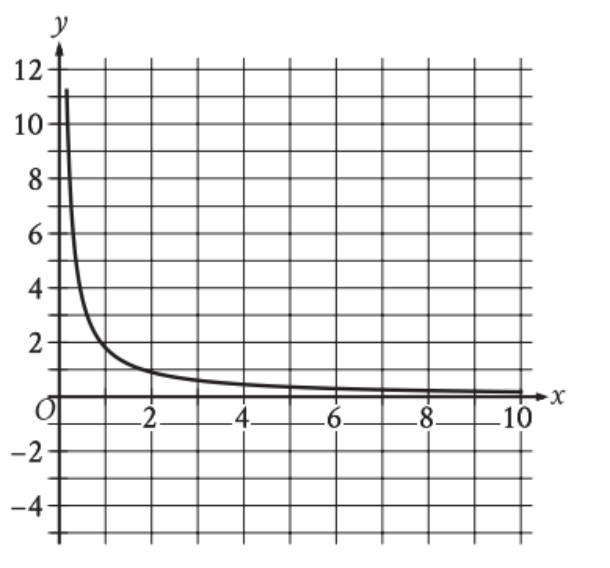
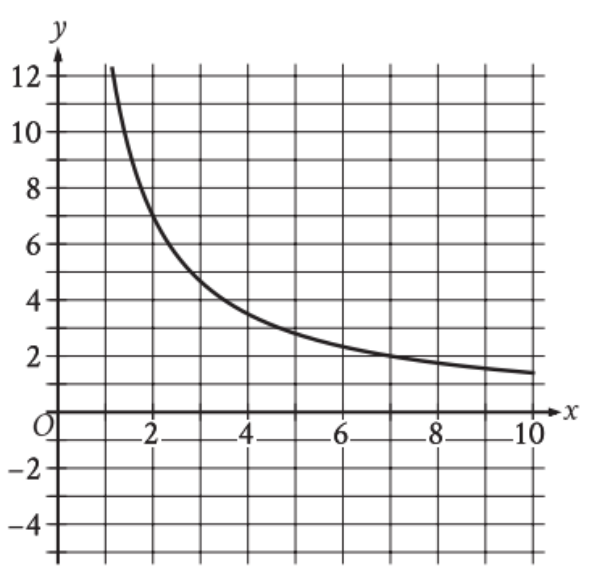
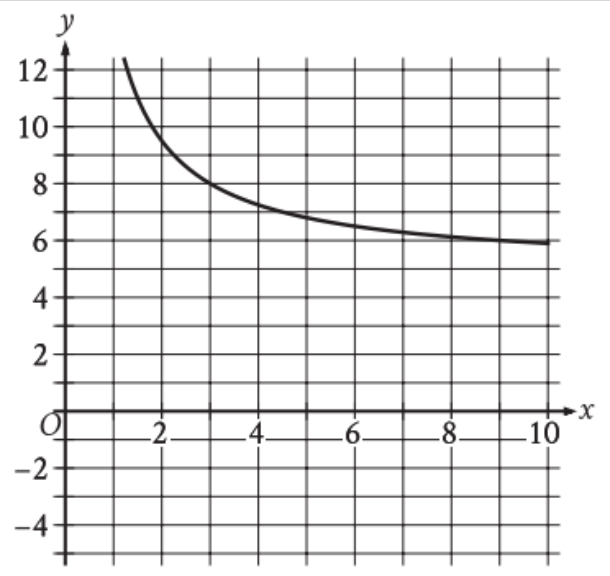
B.



C.



D.



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7	Algebra	Function

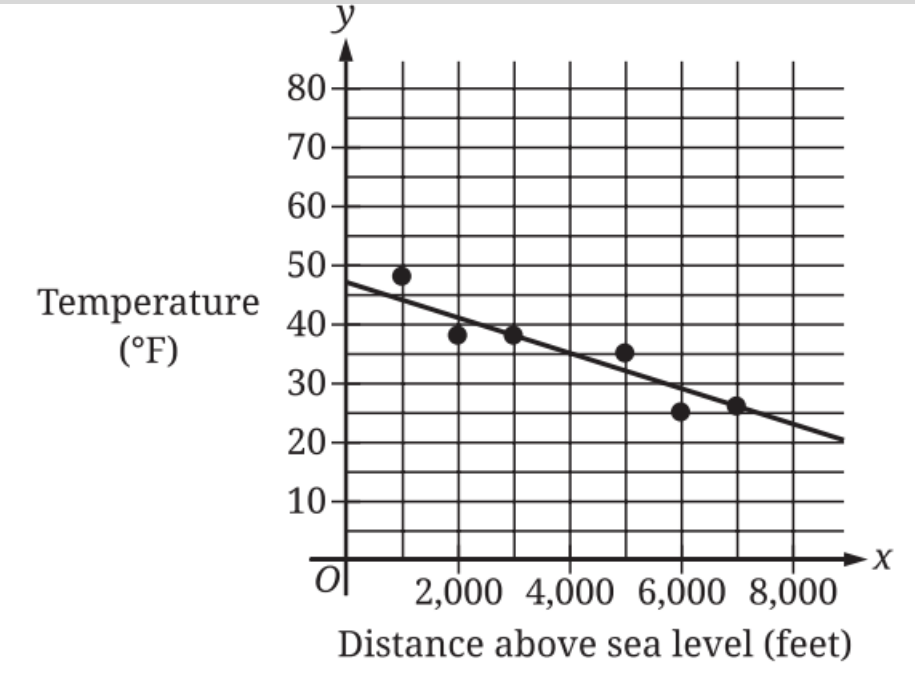
Table A		Table B	
x	$f(x)$	x	$g(x)$
–1	2	3	–1
3	6	5	3
5	5	6	2

Table A shows values that satisfy the function $f(x)$ and Table B shows values that satisfy the function $g(x)$. If $f(g(k)) = 6$, what is the value of k?

A. -1
B. 2
C. 3
D. 5

8	Problem Solving & Data Analysis	Scatterplot

The scatterplot shows the temperature, in degrees Fahrenheit (°F) , and the distance above sea level, in feet, measured at 6 locations on Mount Jefferson. A line of best fit is also shown



At a distance of 4,000 feet above sea level, what is the temperature, in °F , predicted by the line of best fit?

A. 47
B. 35
C. 25
D. 0

0 8 2 1 7 0 2 7 6 0 7 6

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9	Advanced	Radical Expression

$$\frac{12}{n} - \frac{2}{t} = -\frac{2}{w}$$

The given equation relates the variables n, t, and w, where n > 0, t > 0 , and w > t . Which expression is equivalent to n ?

A. $12tw$
B. $6(t - w)$
C. $\frac{w-t}{6tw}$
D. $\frac{6tw}{w-t}$

10	Algebra	Function

The cost of renting a carpet cleaner is \$52 for the first day and \$26 for each additional day. Which of the following functions gives the cost $C(d)$, in dollars, of renting the carpet cleaner for d days, where d is a positive integer?

A. $C(d) = 26d + 26$
B. $C(d) = 26d + 52$
C. $C(d) = 52d - 26$
D. $C(d) = 52d + 78$

11	Problem Solving & Data Analysis	Standard deviation

Class A : 68, 79, 88, 91, 97, 98, 99
Class B : 85, 85, 85, 88, 88, 90, 90

The list above indicate the test score, in increasing order, of two class in an academy. Which of the following correctly compares the standard deviation of the score for each class??

A. The standard deviation of the score in Class A is smaller
B. The standard deviation of the score in Class B is smaller
C. The standard deviation of the score in Class A and class B are equal
D. The relationship cannot be determined from the information given

12	Problem Solving & Data Analysis	Probability

The table summarize the distribution of color and shape for 100 tiles of equal area.

	Red	Blue	Yellow	Total
Square	10	20	25	55
Pentagon	20	10	15	45
Total	30	30	40	100

If one of these tiles is selected at random, what is the probability of selecting a red tile?
(Express your answer as a decimal or fraction, not as a percent.)

13	Advanced Math	Polynomial

Which expression is equivalent to $(x^2 + 11)^2 + (x - 5)(x + 5)$?

A. $x^4 + 23x^2 - 14$
B. $x^4 + 23x^2 + 96$
C. $x^4 + 12x^2 + 121$
D. $x^4 + x^2 + 146$

0 8 2 1 7 0 2 7 6 0 7 6

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14	Advanced Math	Radical	
	What is one of possible solution to the equation $$\frac{6}{x+1} - \frac{3}{x-1} = \frac{1}{4}\,?$$		
15	Advanced Math	Exponent	
	Which expression equivalent to $(m^4q^4z^{-1})(mq^5z^3)$ where $m, q,$ and z are positive? A. $m^4q^{20}z^{-3}$ B. $m^5q^9z^2$ C. $m^6q^8z^{-1}$ D. $m^{20}q^{12}z^{-2}$		
16	Advanced Math	Polynomial	
	What is the coefficient of x^2 when $6x^2 - \frac{2}{5}x + 1$ is multiplied by $10x + \frac{1}{3}$? A. -4 B. -2 C. 2 D. 4		
17	Advanced Math	Quadratic	
	$$f(x) = (x-10)(x+13)$$ The function f is defined by the given equation. For what value of x does $f(x)$ reach its minimum? A. -130 B. -13 C. $-\frac{23}{2}$ D. $-\frac{3}{2}$		
18	Geometry	Area of rectangle	
	Rectangles ABCD and EFGH are similar. The length of each side of EFGH is 6 times the length of the corresponding side of ABCD. The area of ABCD is 54 square units. What is the area, in square units, of EFGH? A. 9 B. 36 C. 324 D. 1,944		
19	Geometry	Area of Triangle + Quadratic Eq	
	The graph equation $y = 2x^2 - 16x + 14$ intersect the y-axis at point A and the x-axis at point B and C. What is the area of triangle ABC? A. 56 B. 54 C. 48 D. 42		

0 8 2 1 7 0 2 7 6 0 7 6

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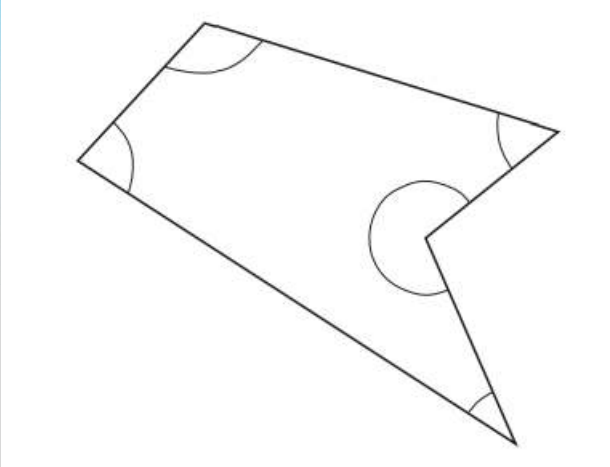
20	Geometry	Angle of Polygon



The figure above show a polygon with five sides. What is the average (arithmetic mean) of the measures, in radian, of the five angle shown?

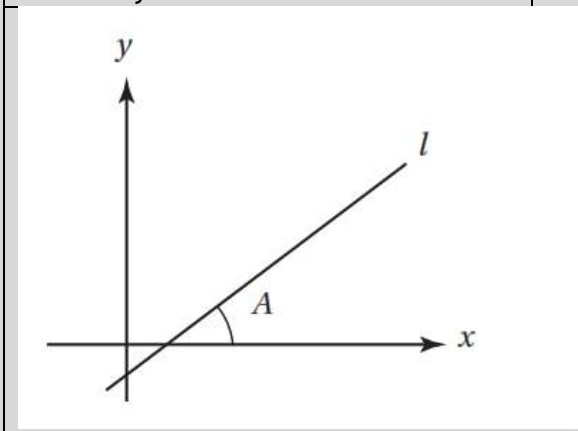
A. $\frac{3}{5}\pi$
B. $\frac{7}{5}\pi$
C. $\frac{8}{7}\pi$
D. π

21	Geometry	3D Figure

A right rectangular prism has a height of 9 inches. The length of the prism's base is x inches, which is 7 inches more than the width of the prism's base. Which function V give the volume of the prism, in cubic inches, in the term of the length of the prism's base?

A. $V(x) = x(x+9)(x+7)$
B. $V(x) = x(x+9)(x-7)$
C. $V(x) = 9x(x+7)$
D. $V(x) = 9x(x-7)$

22	Geometry	Trigonometry



In the figure above, if $\cos A = 0.8$ what is the slope of line l?

A. 0.60
B. 0.75
C. 0.90
D. 1.10

0 8 2 1 7 0 2 7 6 0 7 6

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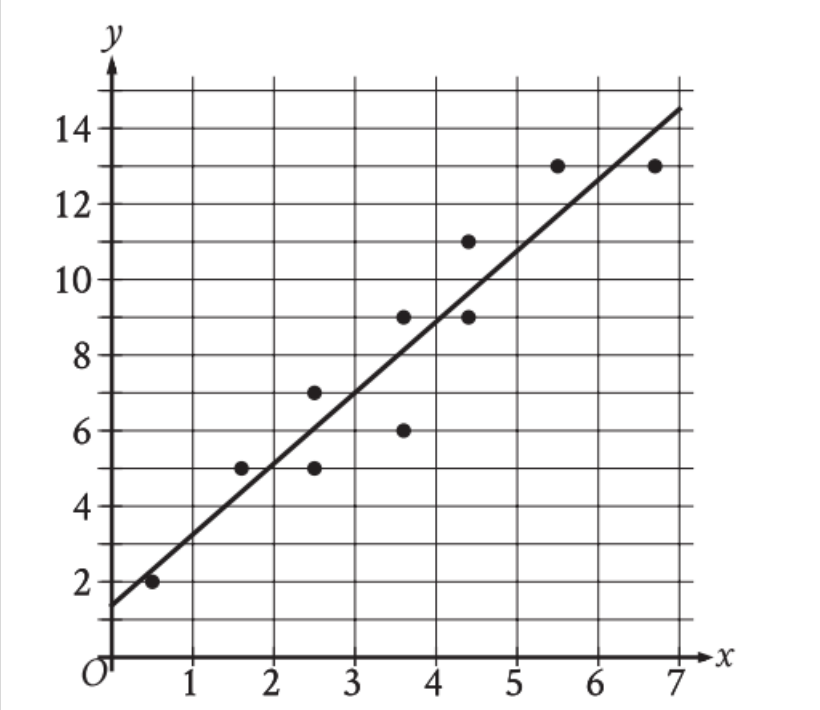
23	Algebra	Linear Equation	
	If $4\sqrt{2x} = 16$, what is the value of 6x? A. 24 B. 48 C. 72 D. 96		
24	Algebra	Linear Equation	
	A line in the xy-plane has a slope of 9 and passes through the point (0, −5) . The equation $y = px + r$ defines the line, where p and r are constants. What is the value of p ?		
25	Problem Solving & Data Analysis	Percent	
	The original price of a motorcycle was marked down by 30% for a week-long sale. Since the item was not sold in the first week, it was marked down an additional p percent. If the total mark down from the original price was 58%, what is the value of p? A. 12 B. 28 C. 40 D. 42		
26	Algebra	Scatterplot	
	In the given scatterplot, a line of best fit for the data is shown.  Which of the following is closest to the slope of the line of best fit shown? A. 0 B. ½ C. 1 D. 2		
27	Problem Solving & Data Analysis	Percent	

0 8 2 1 7 0 2 7 6 0 7 6

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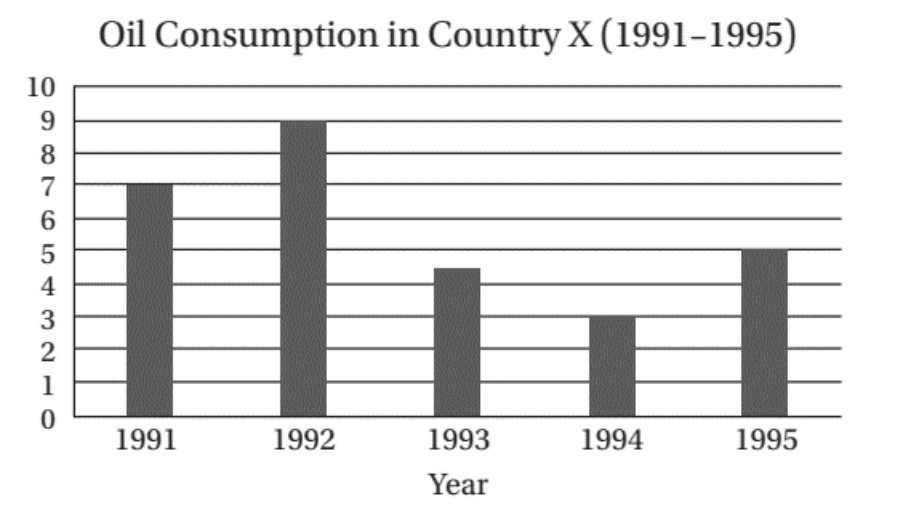
	The amount of Hanna's bill for a food order was \$50. Hanna gave a tip of 20% of the amount of the bill. What is the amount, in dollars, of the tip Hanna gave?		
28	Advanced	Radical	
	Which of the following is equal to $\frac{1}{\sqrt{3}+1}$? A. $\frac{\sqrt{3}}{2}-\frac{1}{2}$ B. $\frac{\sqrt{3}}{2}+\frac{1}{2}$ C. $\frac{\sqrt{3}}{4}-\frac{1}{4}$ D. $\frac{\sqrt{3}}{4}+\frac{1}{4}$		
29	Problem Solving & Data Analysis	Probability	
	At a movie theater, there are a total of 350 customers. Each customer is located in either theater A, theater B, or theater C. If one of these customers is selected at random, the probability of selecting a customer who is located in theater A is 0.48, and the probability of selecting a customer who is located in theater B is 0.24. How many customers are located in theater C? A. 28 B. 40 C. 84 D. 98		
30	Problem Solving & Data Analysis	Histogram	C



The number of barrels of oil consumed by country X each year from 1991 to 1995 is shown in the graph above. If the total amount of oil consumed in those five years was 285.000 barrels, what is an appropriate label for the vertical axis of the graph ?

A. Barrels of oil consumed (in hundred)
B. Barrels of oil consumed (in thousand)
C. Barrels of oil consumed (in tens of thousands)
D. Barrels of oil consumed (in hundreds ofthousands)

0 8 2 1 7 0 2 7 6 0 7 6

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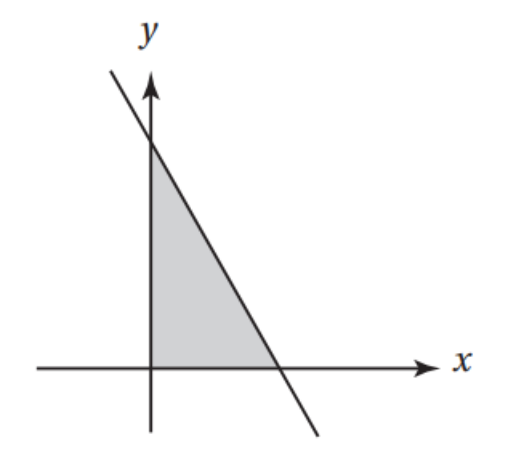
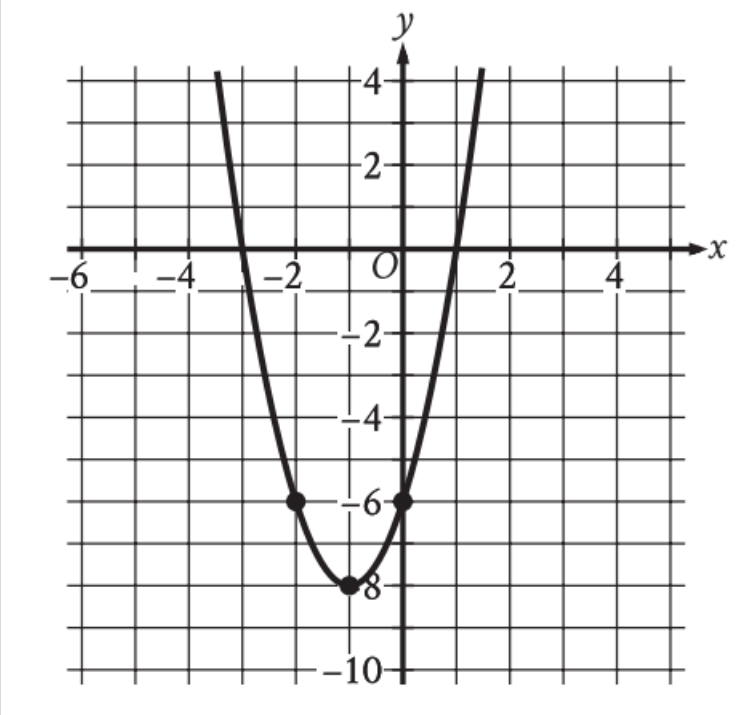
31	Geometry + Algebra	Inequality and Triangle	



$$x \geq 0$$
$$y \geq 0$$
$$3x + y \leq k$$

In the figure above the shaded area region represents the solution set for the system of inequalities shown. If the area of this shaded area is 24 square unit, what is the value of k?

32	Advanced	Quadratic	



The graph of $y = 2x^2 + bx + c$ is shown, where b and c are constants. What is the value of bc?

A. 8
B. -3
C. -8
D. -24

0 8 2 1 7 0 2 7 6 0 7 6

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33	Algebra	Quadratic Function

$$h(t) = -16t^2 + 64t$$

The equation above expresses the approximate height, h , in feet, of a rocket t seconds after it is launched upwards rom the ground until it hits the ground again. After how many seconds will the rocket reach its highest point?

A. 2
B. 4
C. 8
D. 16

34	Algebra	Inequality

$$y < x$$
$$x < 22$$

For which of the following tables are all the values of x and their corresponding values of y solutions to the given system of inequalities?

A.

x	y
19	18
20	19
21	20

B.

x	y
19	20
20	21
21	22

C.

x	y
23	22
24	23
25	24

D.

x	y
23	24
24	25
25	26

0 8 2 1 7 0 2 7 6 0 7 6

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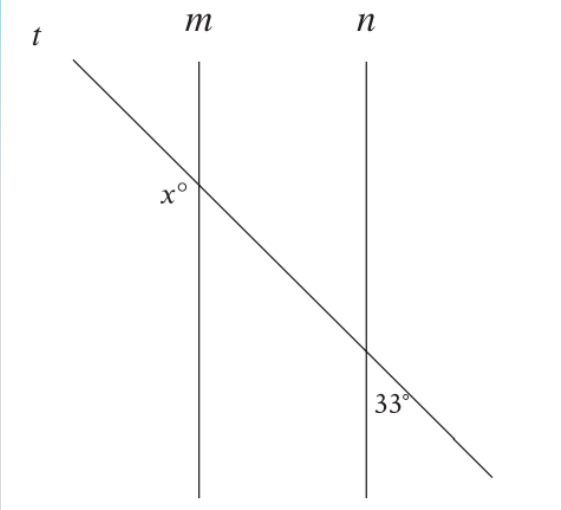
35	Geometry	Angle	



In the figure, line m is parallel to line n, and line t intersects both lines. What is the value of x ?

A. 33
B. 57
C. 123
D. 147

36	Algebra	Functions	

$$f(x) = \frac{x^2 - 36}{\frac{x}{3} - 6}$$

For which value of x is the function above undefined?

A. -6
B. 0
C. 3
D. 18

37	Algebra	Functions	

The function f defined by $f(t) = 14t + 9$ gives the estimated length, in inches, of a vine plant t months after Tyler purchased it. Which of the following is the best interpretation of 9 in this context?

A. Tyler will keep the vine plant for 9 months
B. The vine plant is expected to grow 9 inches each month
C. The vine plant is expected to grow to a maximum length of 9 inches
D. The estimated length of the vine plant was 9 inches when Tyler purchased it

0 8 2 1 7 0 2 7 6 0 7 6

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38 | Problem Solving & Data Analysis | Median

Ages of The First United Stated President at the Beginning of Their Terms in Office

President	Age (years)	President	Age (years)
Washington	57	Jackson	62
Adams	62	Van Buren	55
Jefferson	58	Harrison	68
Madison	58	Tyler	51
Monroe	59	Polk	50
Adams	58	Taylor	65

The table above list the ages of the first 12 United States presidents when they began their term in office. According to the table, what was the median age, in years, of these presidents at the beginning of their terms?

39 | Problem Solving & Data Analysis | Mean

Five *Eretmochelys imbricata*, a type of sea turtle, each have a nest. The table shows an original data set of the number of eggs that each turtle laid in its nest.

Nest	Number of eggs
A	149
B	144
C	148
D	136
E	139

A sixth nest with 121 eggs is added to create a new data set. Which of the following correctly compares the means of the two data sets?

A. The mean of the original data set is greater than the mean of the new data set.
B. The mean of the original data set is less than the mean of the new data set.
C. The means of both data sets are equal.
D. There is not enough information to compare the means.

40 | Advanced | Rational Expression

$$\frac{4x^2+1}{4x^2-1}+\frac{16x}{16x^2-4}$$

The expression above is equivalent to $\frac{ax+b}{ax-b}$ where a and b are constant and $x \neq \frac{1}{2}$. What is the value of $a+b$?

41 | Geometry | Trigonometry

In triangle XYZ, angle Z is a right angle and the length of YZ is 24 units. If $\tan X = \frac{12}{35}$, what is the perimeter , in units, of triangle XYZ?

A. 188
B. 168
C. 84
D. 71

0 8 2 1 7 0 2 7 6 0 7 6

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42	Algebra	System of Equation	
	$(x-2)-4(y+7)=117$ $(x-2)+4(y+7)=442$ The solution to the given system oof equation is (x,y). What is the value of $6(x-2)$?		
43	Geometry	Circle	
	A circle in the xy −plane has the equation $(x-13)^2+(y-k)^2=64$. Which of the following gives the center of the circle and its radius? A. The center is at $(13,k)$ and the radius is 8 B. The center is at $(k,13)$ and the radius is 8 C. The center is at $(k,13)$ and the radius is 64 D. The center is at $(13,k)$ and the radius is 64		
44	Geometry	Volume	
	A gardener has three right cylindrical flower pots, each with a diameter of 16 cm, so that he will fill the potting soil. If her bag of potting soil contains $1{,}000\ in^3$ of soil and he fill all three flower pots to a height of 20 cm, approximately how many cubic inches of soil will be left in the bag? $(1cm^3=0.061\ in^3)$ A. 64 B. 264 C. 736 D. 755		

0 8 2 1 7 0 2 7 6 0 7 6

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